

Module 3

Syllabus - Linear Opamp Circuits: Summing Amplifier and D/A Converter, [Nonlinear Op-amp Circuits: Comparator with zero reference, Comparator with non-zero references. Comparator with Hysteresis. Oscillator: Theory of Sinusoidal Oscillation, The Wein-Bridge Oscillator, RC Phase Shift Oscillator, The Colpitts Oscillator, Hartley Oscillator, Crystal Oscillator. The 555 timer: Monostable Operation, Astable Operation.

Q1. With a neat circuit diagram and equations, explain the operation of two input inverting Summing Amplifier.

Whenever it is required to combine two or more analog signals into a single output, the Summing amplifier is a natural choice.

Figure shows the circuit diagram of an inverting summing amplifier where V_1 and V_2 are inputs applied to the inverting terminal of the op-amp.

Applying KCL to node A $\rightarrow I_1 + I_2 + I_f = 0$

$$\frac{V_A - V_1}{R_1} + \frac{V_A - V_2}{R_2} + \frac{V_A - V_o}{R_f} = 0$$

Due virtual short between the input terminals of an op-amp, $V_A = V_B = 0$

$$\frac{0 - V_1}{R_1} + \frac{0 - V_2}{R_2} + \frac{0 - V_o}{R_f} = 0$$

$$\text{The amplified summer output } V_o = - \left(\left(\frac{R_f}{R_1} \right) V_1 + \left(\frac{R_f}{R_2} \right) V_2 \right) = -[A_{V1}V_1 + A_{V2}V_2]$$

$$\text{Where the gains } A_{V1} = - \left(\frac{R_f}{R_1} \right) \text{ and } A_{V2} = - \left(\frac{R_f}{R_2} \right)$$

$$\text{For } R_f = R_1 = R_2 \rightarrow V_o = - (V_1 + V_2) \text{ ----- Summer}$$

$\rightarrow$ Hence the output is the negative sum of inputs $\leftarrow$

$$\text{If } R_1 = R_2 = 2 R_f \rightarrow V_o = - \left(\frac{R_f}{2R_f} V_1 + \frac{R_f}{2R_f} V_2 \right) = - \left(\frac{V_1 + V_2}{2} \right) \text{ ----- Averaging circuit}$$

If the levels of the inputs and gain can be controlled by using adjustable resistors instead of fixed ones, the circuit works as a Mixer for a high fidelity audio system to combine audio signals.

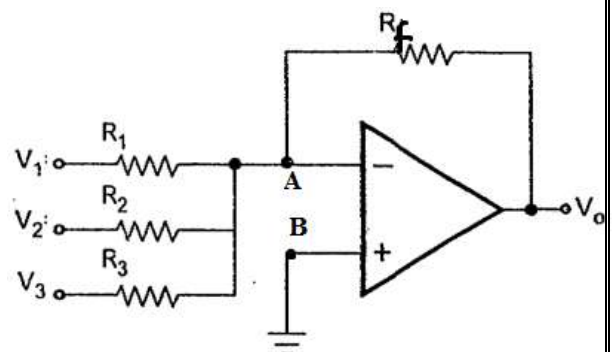
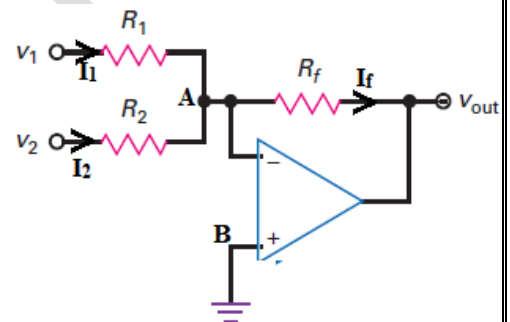
Q2. Explain the operation of three input Inverting Summer and Averaging Circuit

Applying KCL to node A

$$\rightarrow I_1 + I_2 + I_3 + I_f = 0$$

$$\frac{V_A - V_1}{R_1} + \frac{V_A - V_2}{R_2} + \frac{V_A - V_3}{R_3} + \frac{V_A - V_o}{R_f} = 0$$

Due virtual short between the input terminals of an op-amp, $V_A = V_B = 0$



$$\frac{0-V_1}{R_1} + \frac{0-V_2}{R_2} + \frac{0-V_3}{R_3} + \frac{0-V_0}{R_f} = 0$$

$$\text{or } V_0 = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)$$

For $R_f = R_1 = R_2 = R_3 \rightarrow V_0 = -(V_1 + V_2 + V_3)$

$\rightarrow$ Hence the output is the negative sum of inputs $\leftarrow$

If $R_1 = R_2 = R_3 = 3 R_f \rightarrow V_0 = -\left(\frac{R_f}{3R_f} V_1 + \frac{R_f}{3R_f} V_2 + \frac{R_f}{3R_f} V_3\right) = -\left(\frac{V_1+V_2+V_3}{3}\right)$ ----- Averaging circuit

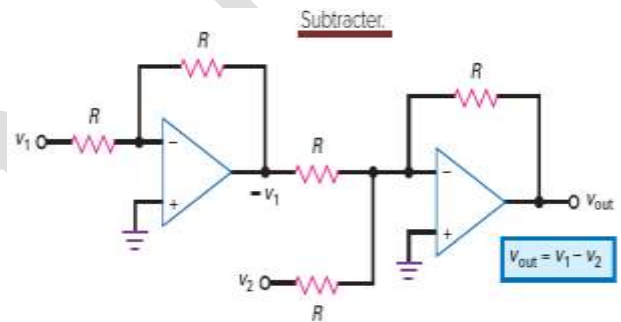
Q3. Explain the working of two input op-amp Subtractor circuit

Figure shows a circuit that subtracts two input voltages to produce an output voltage equal to the difference of inputs v_1 and v_2 . The first op-amp is an inverting amplifier with unity gain driven by the input v_1 . The output of the first stage is $v_{o1} = -A_{v1} v_1 = -(1)v_1 = -v_1$

This voltage is one of the inputs to the second-stage summing circuit. The other input is v_2 .

Since the gain of each channel is unity, the final

output voltage $v_{out} = -A_{v2}[-v_1 + v_2] = -(1)[-v_1 + v_2] = [v_1 - v_2]$



Q4. Explain the operation of Summing amplifier which has both inverting and non-inverting inputs with equations and show that

$$V_o = V_{o1} + V_{o2} = V_3 + V_4 - V_1 - V_2$$

Figure shows a summing amplifier which has both inverting and non-inverting inputs. The inverting side of the amplifier has two input channels, and the non-inverting side has two input channels. The total gain is the superposition of the channel gains and can be obtained by adding the individual gains considering the effect of one input at a time.

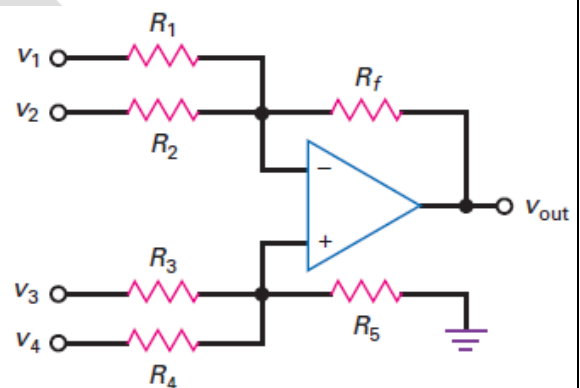
(i) When V_1 and V_2 are considered and other two inputs are set to 0. The circuit works as a two input inverting summer

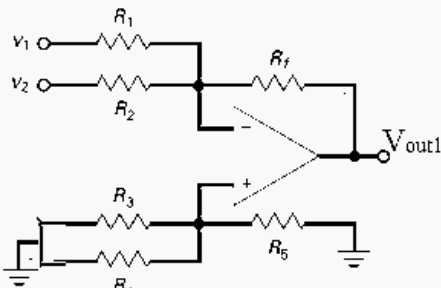
$$V_{o1} = \left(\left(-\frac{R_f}{R_1}\right) V_1 + \left(-\frac{R_f}{R_2}\right) V_2\right)$$

(ii) When only V_3 is considered and other three inputs are set to 0. The circuit works as a non-inverting amplifier.

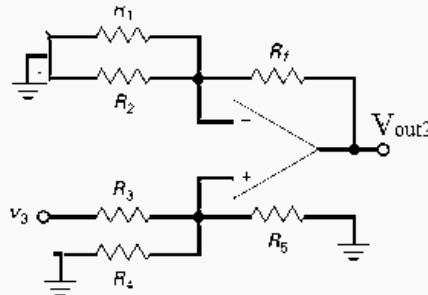
The input applied to the non-inverting terminal is $v^1_3 = \left[\frac{R_4 || R_5}{R_3 + R_4 || R_5}\right] V_3$

$$V_{out2} = \left[1 + \frac{R_f}{R_1 || R_2}\right] v^1_3 = \left[1 + \frac{R_f}{R_1 || R_2}\right] \left[\frac{R_4 || R_5}{R_3 + R_4 || R_5}\right] V_3$$

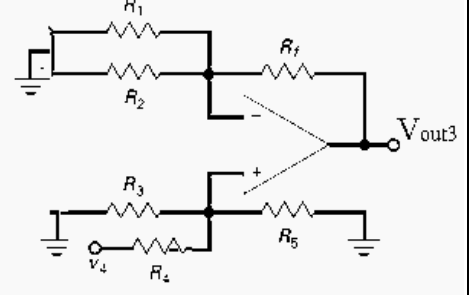




when only V_1 and V_2 are acting and other inputs are zero



when only V_3 is acting and other inputs are zero



when only V_4 is acting and other inputs are zero

(iii) When only V_4 is considered and other three inputs are set to 0. The circuit works as a non-inverting amplifier.

The input applied to the non-inverting terminal is $v^1_4 = \left[\frac{R_3 || R_5}{R_4 + R_3 || R_5} \right] V_4$

$$V_{out2} = \left[1 + \frac{R_f}{R_1 || R_2} \right] v^1_4 = \left[1 + \frac{R_f}{R_1 || R_2} \right] \left[\frac{R_3 || R_5}{R_4 + R_3 || R_5} \right] V_4$$

$$V_{out} = V_{out1} + V_{out2} + V_{out3}$$

$$V_{out} = \left(\left(-\frac{R_f}{R_1} \right) V_1 + \left(-\frac{R_f}{R_2} \right) V_2 \right) + \left[1 + \frac{R_f}{R_1 || R_2} \right] \left[\frac{R_4 || R_5}{R_3 + R_4 || R_5} \right] V_3 + \left[1 + \frac{R_f}{R_1 || R_2} \right] \left[\frac{R_3 || R_5}{R_4 + R_3 || R_5} \right] V_4$$

If all resistors of equal value, $R_1 = R_2 = R_3 = R_4 = R_5 = R_f$

$$V_{out} = ((-1) V_1 + (-1) V_2) + \left[1 + \frac{1}{0.5} \right] \left[\frac{0.5}{1+0.5} \right] V_3 + \left[1 + \frac{1}{0.5} \right] \left[\frac{0.5}{1+0.5} \right] V_4$$

$$V_o = V_{O1} + V_{O2} = V_3 + V_4 - V_1 - V_2$$

Digital to Analog Converters (DAC)

Introduction - The real world physical quantities such as voltage, current, temperature, pressure, time etc are available in analog form, and it is difficult to process, store and transmit analog signals without considerable error because of the noise present in the system. For processing, transmission and storage purposes, it is convenient to express these quantities in digital form. Digital systems have good noise immunity and better accuracy and supports storage, editing and any time retrieval. The operation of any digital communication system is based upon analog to digital (A/D) and digital to analog (D/A) conversion.

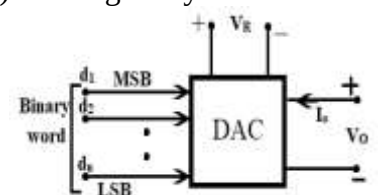
Basic DAC Techniques:

A Digital to Analog Converter (DAC) converts the binary output (digital output) of a digital system into some equivalent analog voltage or current.

A DAC accepts an n bit binary word D whose digits are $[D = d_1, d_2, \dots, d_n]$ and is combined with a reference voltage V_R to give a proportional analog output signal. The output of the DAC can be either a voltage or current.

There are mainly two techniques used for analog to digital conversion

(i) Binary weighted resistor D/A converter (ii) R / 2R Ladder D/A converter



Q5(a). With a neat circuit diagram, explain the working of Binary weighted resistor DAC. What are its limitations.

Figure shows the circuit diagram of a 4 bit binary weighted resistor DAC and its output table. The DAC uses an inverting summing op-amp with a binary weighted resistor 2^0R , 2^1R , 2^2R , 2^3R - - - 2^nR . The input voltages are digital or two-state, which means that they can be either 0V (logic 0) or V_{ref} (logic 1)

The output voltage for a n bit binary weighted DAC can be expressed as

$$V_{out} = -\left(\frac{1}{2^n}\right) [V_{n-1}2^n + V_{n-2}2^{n-1} + V_{n-3}2^{n-2} + \dots]$$

For MSB the gain is 1 and for LSB the gain is $\frac{1}{2^{n-1}}$ Where n indicates the number of bits or inputs.

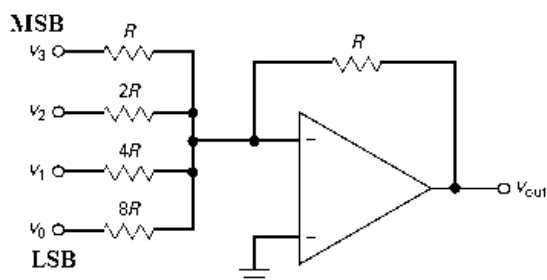
For 4 bit DAC (n = 4), the output can be expressed as

$$V_{out} = -\left(\frac{1}{2^3}\right) [V_{4-1}2^4 + V_{4-2}2^{3-1} + V_{4-3}2^{4-2} + V_{4-4}2^{4-3}] = -[V_32^0 + V_22^{-1} + V_12^{-2} + V_02^{-3}]$$

$$V_{out} = -[V_3 + 0.5V_2 + 0.25V_1 + 0.125V_0]$$

When all the inputs are high (logic 1) the output will be

$$V_{out} = -[V_{ref} + 0.5V_{ref} + 0.25V_{ref} + 0.125V_{ref}] = -1.875 V_{ref}$$



$$V_{out} = -(V_3 + 0.5V_2 + 0.25V_1 + 0.125V_0)$$

Digital inputs				Analog o/p
V_3	V_2	V_1	V_0	
0	0	0	0	0
0	0	0	1	$-0.125V_{ref}$
0	0	1	0	$-0.25 V_{ref}$
.	.	.	.	.
.	.	.	.	.
1	1	1	0	$-1.75 V_{ref}$
1	1	1	1	$-1.875 V_{ref}$

The output is a stair- case signal which is passed through a Lowpass filter to generate the equivalent analog signal.

Disadvantages of Binary Weighted Resistor scheme:

The binary-weighted D/A converter can be used in applications where the number of inputs is limited and where high precision is not required. When a higher number of inputs is used, a higher number of different resistor values is required which affects the accuracy and stability of the D/A converter. For an 8 bit DAC, the largest resistor is 128 times the smallest one. Due to this there exists a limitation on the selection values for resistors, which in turn affects the accuracy and causes loading effect.

Q5(b). With a neat circuit diagram, explain the working of R/2R Ladder DAC. What are its advantages.

In R-2R ladder DAC, resistors of only two values namely R and 2R are used; hence it is more suitable for IC fabrication. Typical values of R ranges from 2.5k Ω to 10k Ω .

The resistors are so arranged to form a ladder network and the output voltage is a weighted sum of digital inputs. Figure shows the circuit of a 4-bit R/2R ladder D/A converter. The switches $D_0 - D_3$ would normally be some type of active switch. The switches connect the four inputs to either ground (logic 0) or $+V_{ref}$ (logic 1).

The ladder network converts the possible binary input values from 0000 through 1111 to one of 16 unique output voltage levels as shown in the table. The output of the DAC can be expressed using the relation.

$$V_{out} = -\left(\frac{BIN}{2^{n-1}}\right) V_{ref} \quad \text{where } BIN = [D_0 2^0 + D_1 2^1 + D_2 2^2 + D_3 2^3]$$

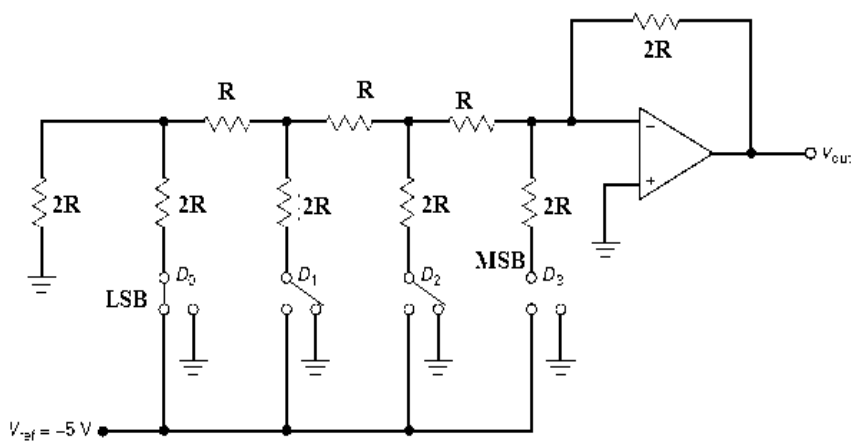
For $[D_3 D_2 D_1 D_0] = [1 0 0 1] \rightarrow BIN = [D_0 2^0 + D_1 2^1 + D_2 2^2 + D_3 2^3] = [2^0 + 0 + 0 + 2^3] = 9$

$$V_{out} = -\left(\frac{BIN}{2^{n-1}}\right) V_{ref} = -\left(\frac{9}{2^{4-1}}\right) 5 = -5.625V$$

Similarly for $[D_3 D_2 D_1 D_0] = [1 1 0 1]$

$$\rightarrow BIN = [D_0 2^0 + D_1 2^1 + D_2 2^2 + D_3 2^3] = [2^0 + 0 + 2^2 + 2^3] = 13$$

$$V_{out} = -\left(\frac{BIN}{2^{n-1}}\right) V_{ref} = -\left(\frac{13}{2^{4-1}}\right) 5 = -8.125 V$$



Digital inputs				Analog o/p
D ₃	D ₂	D ₁	D ₀	In volts
0	0	0	0	0
0	0	0	1	-- 0.625
0	0	1	0	--1.25
.	.	.	.	.
.	.	.	.	.
1	1	1	0	--8.75
1	1	1	1	--9.375

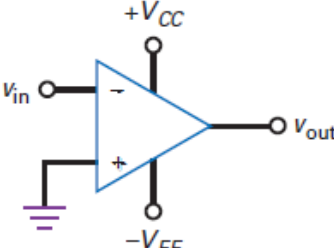
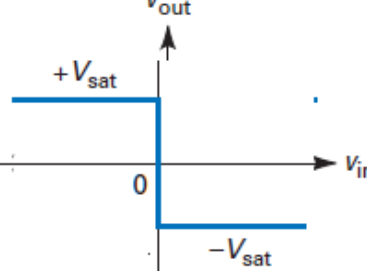
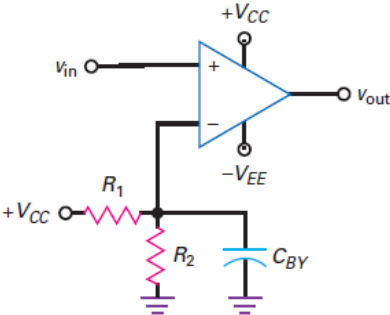
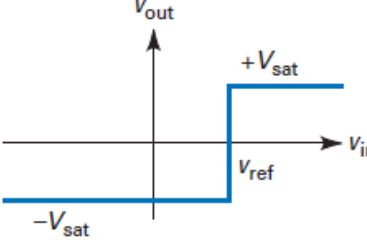
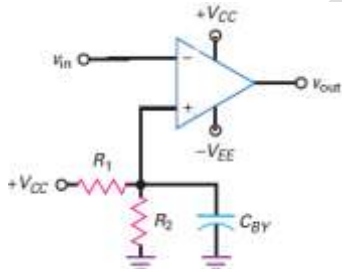
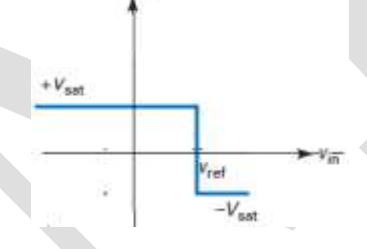
Advantages - The accuracy and stability of the D/A converter depends on the absolute accuracy of the resistors and their ability to track each other with temperature variations. In R-2R ladder DAC, only two resistor values are required, this method lends itself to ICs with 8-bit or higher binary inputs and provides a higher degree of accuracy

Comparators

Q7 What is Comparator? Explain the different types of Comparators without and with reference.

A Comparator compares the input signal voltage applied to one input terminal of op-amp with a known voltage called as reference voltage applied to the other input terminal. In its simplest form, it is an open loop amplifier with two analog inputs and a digital output which can be either $+V_{sat}$ depending on which input is larger

Comparator type	Transfer characteristics	Remarks
Non-inverting Comparator with zero reference 		In non-inverting comparator the input V_{in} is applied to the non-inverting terminal and zero voltage is applied to the inverting terminal. Operation For $V_{in} < 0 \rightarrow V_O = -V_{sat}$ For $V_{in} > 0 \rightarrow V_O = +V_{sat}$ <u>A comparator with zero reference is also called as zero cross detector (ZCD)</u>

Inverting Comparator with zero reference 		In inverting comparator the input V_{in} is applied to the inverting terminal and a zero voltage is applied to the non-inverting terminal. Operation For $V_{in} < 0 \rightarrow V_O = +V_{sat}$ $V_{in} > 0 \rightarrow V_O = -V_{sat}$
Non-inverting Comparator with positive reference voltage 	 $V_{ref} = \frac{+V_{CC}R_2}{R_1 + R_2}$ The reference become negative if $-V_{EE}$ is used.	In non-inverting comparator the input V_{in} is applied to the non-inverting terminal and a fixed reference voltage V_{ref} is applied to the inverting terminal. The reference voltage is obtained from the voltage divider network. Operation For $V_{in} < V_{ref}, \rightarrow V_O = -V_{sat}$ $V_{in} > V_{ref}, \rightarrow V_O = +V_{sat}$
A bypass capacitor reduces the amount of power-supply ripple and other noise appearing at the inverting input. The cutoff frequency of the bypass circuit is given by : $f_c = \left[\frac{1}{2\pi(R_1 R_2)C_{BY}} \right] \text{ Hz}$		
Inverting Comparator with positive reference voltage 	 $V_{ref} = \frac{+V_{CC}R_2}{R_1 + R_2}$ The reference become negative if $-V_{EE}$ is used.	In inverting comparator the input V_{in} is applied to the non-inverting terminal and a fixed reference voltage V_{ref} is applied to the non-inverting terminal. The reference voltage is obtained from the voltage divider network. Operation For $V_{in} < V_{ref}, \rightarrow V_O = +V_{sat}$ $V_{in} > V_{ref}, \rightarrow V_O = -V_{sat}$
A bypass capacitor reduces the amount of power-supply ripple and other noise appearing at the non-inverting input. The cutoff frequency of the bypass circuit is given by : $f_c = \left[\frac{1}{2\pi(R_1 R_2)C_{BY}} \right] \text{ Hz}$		

Comparator with Hysteresis - Schmitt Trigger (Regenerative Comparator)

Q8 What is a regenerative comparator? With a neat circuit diagram and equations, explain inverting Schmitt trigger.

Need - The basic comparators are used in open loop mode and since the open loop gain of op-amp is very high, very small noise voltages can also cause false triggering at the output. The comparator circuit designed with a positive feedback to avoid such an unwanted false triggering is called Schmitt Trigger or Regenerative Comparator.

Figure shows the circuit diagram of inverting Schmitt trigger with input-output waveforms and transfer curve.

The input is applied to the inverting terminal of the Schmitt trigger and the feedback signal is given to the non-inverting terminal. This ensures positive feedback. The reference voltages are obtained using voltage divider network $R_1 - R_2$.

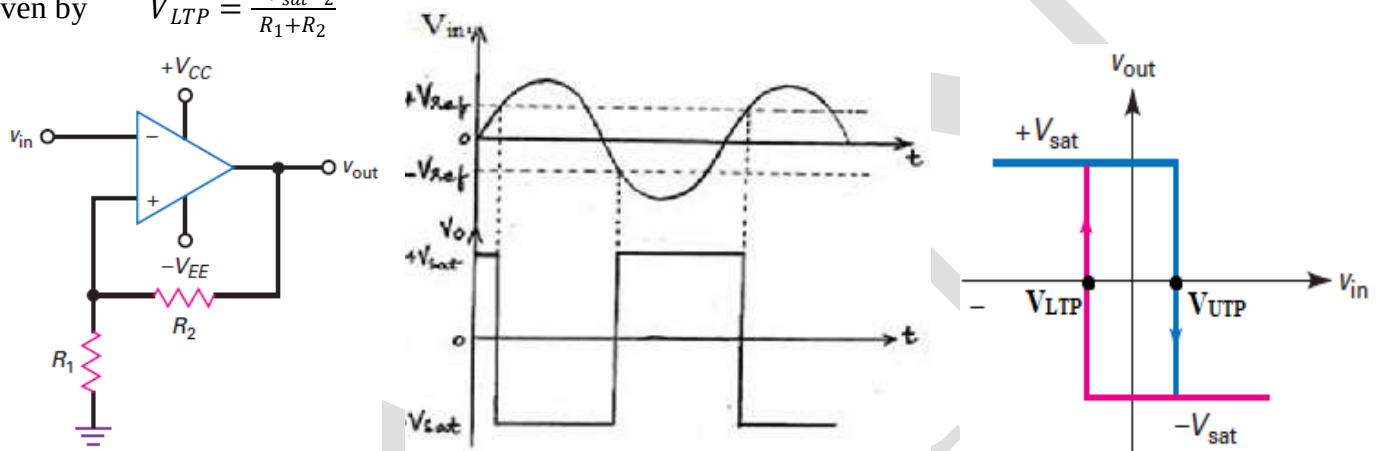
The reference voltage at the non-inverting input terminal is given by $V_{ref} = \frac{V_o R_2}{R_1 + R_2}$

When the output voltage is at $+V_{sat}$, then the reference voltage is called as Upper Threshold Potential UTP

$$V_{UTP} = \frac{+V_{sat} R_2}{R_1 + R_2}$$

When $V_{in} > V_{ref}$ or V_{UTP} , then the voltage at inverting terminal becomes greater than non inv terminal and the output switches to $-V_{sat}$ level and now the new reference voltage is called Lower Triggering Potential LTP and

is given by $V_{LTP} = \frac{-V_{sat} R_2}{R_1 + R_2}$



By ensuring the threshold voltages are larger than noise voltages, false output triggering can be eliminated.

The input voltage value at which the output changes from $+V_{sat}$ to $-V_{sat}$ is called Upper Threshold Potential (UTP). The value of the input at which the out changes from $-V_{sat}$ to $+V_{sat}$ is Lower Threshold Point (LTP).

The voltage difference between V_{UTP} and V_{LTP} is referred as Hysteresis and is given by

$$V_H = [V_{UTP} - V_{LTP}]$$

Questions from previous question papers

DEC23-Jan2024

- 1). Explain how an op-amp summer circuit be configured to function as a subtractor. (5M)
- 2). How does the design and configuration of an op-amp R-2R DAC contribute to its accuracy and performance in converting digital signals to analog signals? (8M)
- 3). Describe the working of inverting Schmitt trigger circuit. How is Schmitt trigger circuit different from regular comparator circuit. Explain with the help of Hysteresis curve. (8M)
- 4). Explain the operation of Colpitt's Oscillator with CE configuration. (6M)
- 5). Explain the operation of Monostable operation of 555 timer. (6M)

June-July 2024

- 1). Derive V_{ref} and f_c of comparators with non zero reference. (10M)
- 2). With neat diagram, explain R-2R DAC and derive the expression for V_{out} . (10M)
- 3). Explain the operation of RC phase shift Oscillator. (10M)
- 4). With a neat diagram, explain the operational amplifier based Wien bridge Oscillator circuit. (10M)

Module 3(b)

Syllabus -. Oscillator: Theory of Sinusoidal Oscillation, The Wein-Bridge Oscillator, RC Phase Shift Oscillator, The Colpitts Oscillator, Hartley Oscillator, Crystal Oscillator. The 555 timer: Monostable Operation, Astable Operation.

Oscillators and Principle of Operation of Sinusoidal Oscillator:

Q9. What is an Oscillator ? Explain the principle of sinusoidal oscillators.

An Oscillator is an electronic device which produces sustained sinusoidal oscillations of desired frequency. An Oscillator consists of an amplifier, a positive feedback circuit and tank circuit. Usually the feedback circuit behaves as the tank circuit.

Consider an amplifier with gain A_v driven by an input voltage source v_{in} . The amplified output voltage v_{out} drives a feedback circuit that is usually a resonant circuit. $v_{out} = A_v v_{in}$

The feedback voltage returning to point x is given by:

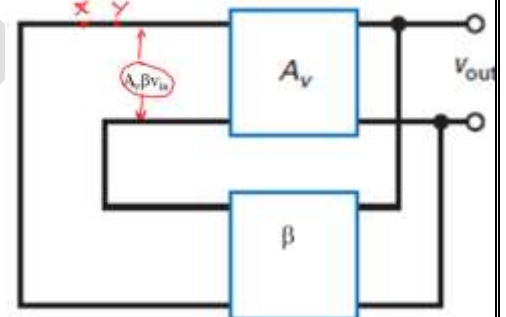
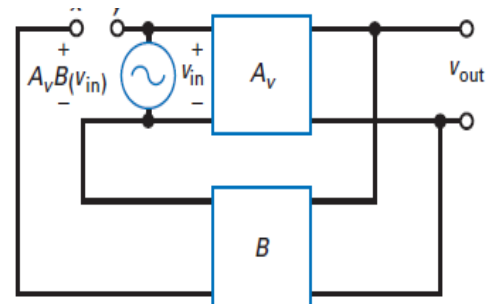
$$v_f = \beta v_{out} = A_v \beta v_{in}$$

where β is the feedback fraction and where βA_v is referred to as the loop gain. If the phase shift through the amplifier and feedback circuit is equivalent to 0° , $A_v \beta (v_{in})$ is in phase with v_{in} .

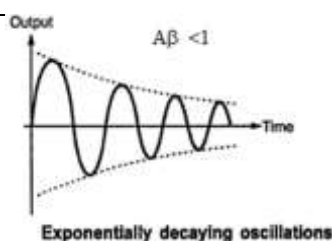
Hence the output voltage $v_{out} = A_v[v_{in} + v_f] = A_v[v_{in} + A_v \beta v_{in}]$

Suppose the point x and point y are connected and simultaneously the voltage source v_{in} is removed, then the feedback voltage $A_v \beta v_{in}$ drives the input of the amplifier and the circuit will continue operating. The output waveform will still exist after the switch is closed if the condition $\beta A_v \geq 1$ is met. $v_{out} = A_v[v_{in} + v_f] = A_v[v_{in} + \beta v_{out}]$ or $v_{out}(1 - \beta A_v) = A_v v_{in}$

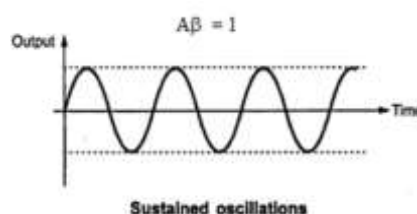
$\frac{v_{out}}{v_{in}} = \frac{A_v}{(1 - \beta A_v)}$ If $v_{in} = 0$, the only way that v_{out} can be non zero is that loop gain $\beta A_v = 1$ which implies that the equation becomes indeterminate (0/0).



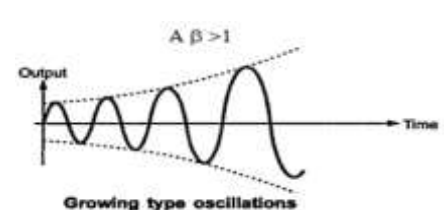
If $A_v \beta < 1$, then $A_v \beta (v_{in}) < v_{in}$ and the output signal will gradually die out resulting in damped oscillations



If $A_v \beta = 1$, then $A_v \beta (v_{in}) = v_{in}$ and the output voltage is a steady, resulting in sustained sinusoidal oscillations



If $A_v \beta > 1$ then $A_v \beta (v_{in}) > v_{in}$ and the output voltage builds up resulting in growing oscillations.



Barkhausen's Criterion (Conditions for Sustained Oscillations)

Q10. Mention the conditions for required for Sustained Oscillations

- (i) The total phase shift around the amplifier and feedback network $|A\beta|$ is 0° or 360° .
- (ii) The magnitude of the product of open loop gain A of the amplifier and magnitude of the feedback factor β is unity i.e. $|A\beta| = 1$

Note: Initially $|A\beta|$ must be kept >1 to start the oscillations and later the circuit adjusts itself in such a way that $|A\beta|=1$, so that sustained oscillations are produced. Figure below shows the oscillations for different values of $A\beta$

Applications of Oscillators

Oscillators are used in radio receivers to transform the RF signals to IF signals

Oscillators are used generate RF carrier signal in a radio transmitters

Oscillators are used to generate clock signal for digital systems;

Oscillators are used to generate sweep signals used in TV and CRO.

Q11. Write short notes on (a) Lag circuit (b) Lead circuit.

A Lag circuit is RC bypass circuit in which the output is taken across the capacitor and the output voltage lags behind the input voltage by a phase angle Φ as shown in the figure.

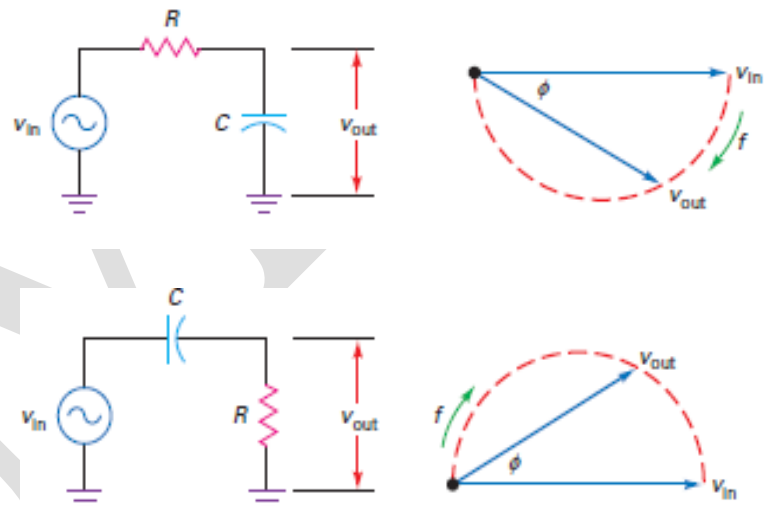
The voltage gain $\frac{V_{out}}{V_{in}} = \frac{X_C}{\sqrt{R^2 + X_C^2}}$ and the

Phase angle $\phi = -\tan^{-1}\left(\frac{R}{X_C}\right)$

A Lead circuit is RC coupling circuit in which the output is taken across the resistor and the output voltage leads the input voltage by a phase angle Φ as shown in the figure.

The voltage gain $\frac{V_{out}}{V_{in}} = \frac{R}{\sqrt{R^2 + X_C^2}}$ and the Phase

angle $\phi = \tan^{-1}\left(\frac{X_C}{R}\right)$

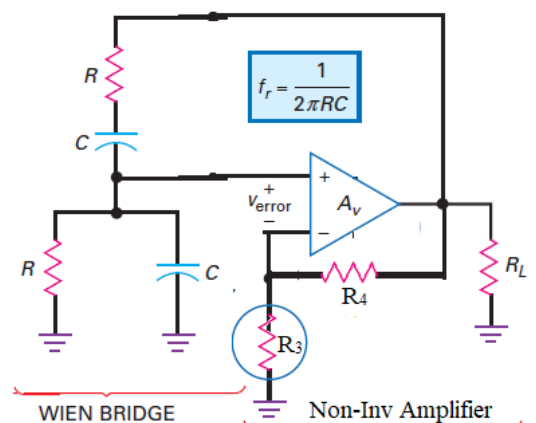


The Wien-Bridge Oscillator

Q12. With a neat circuit diagram, explain the operation of Wien bridge oscillator

The **Wien-bridge Oscillator** is the standard oscillator circuit for low to moderate frequencies in the range of **5 Hz to about 1 MHz**. It is often used in commercial audio generators and is usually preferred for other low-frequency applications.

Construction - A Wien bridge is an ac bridge which has two arms as one RC series (Lag circuit) network and another RC shunt (Lead circuit) network. The lead-lag circuit forms the resonant feedback circuit. The output of the bridge is applied to a non inverting



amplifier and the output of the amplifier is applied as input to the bridge. The resistors R_3 and R_4 of the op-amp forms a part of the bridge (other two arms of the bridge). The feedback voltage is attenuated but not phase shifted by the feedback network, the feedback voltage is amplified to produce oscillations.

Operation - At low frequencies, the series capacitor acts as open circuit and no input is applied and at high frequencies, the shunt capacitor acts as short circuit, again no input is applied. Hence the output is zero in both the cases.

In between these two extremes, at one particular frequency, the bridge gets balanced and sustained oscillations are produced. This frequency is called resonant frequency of the bridge.

From the analysis of the bridge, balance is obtained when the following equations are fulfilled.

$$\frac{R_4}{R_3} = 2 \quad \text{The gain of the amplifier } A_v \geq 1 + \frac{R_4}{R_3} = 3$$

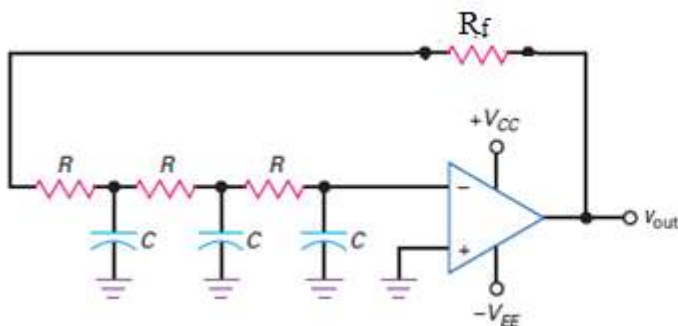
$$\text{The frequency of Oscillation } f_r = \frac{1}{2\pi RC} \text{ Hz}$$

RC Phase Shift Oscillator

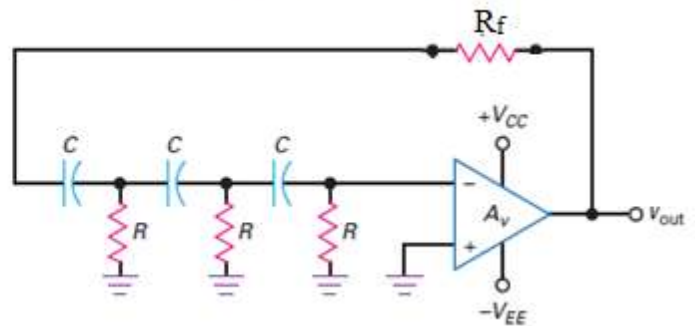
Q13. Explain the operation of RC phase shift oscillator.

A RC phase shift oscillator basically consists of an amplifier and a ladder type feedback network consisting of resistors and capacitors. The phase-shift oscillator with three lead circuits (or lag circuits) in the feedback path. A lead circuit produces a phase shift between 0° and 90° . Using proper values of R and C, the phase angle ϕ is adjusted to 60° .

Phase-shift oscillator with three lag circuits.



Phase-shift oscillator with three Lead circuits.



A RC ladder network is used in the feedback. By cascading 3 similar RC sections each giving a phase angle of 60° , a total phase shift of 180° can be obtained. The inverting amplifier gives 180° phase shift, hence the total phase shift around the loop is 0° or 360° . The feedback network gives an attenuation of $(1/29)$, hence the gain of the amplifier A_v must be greater than 29 for generating sustained oscillations.

$$\text{The frequency of oscillations are given by } f = \frac{1}{2\pi\sqrt{6} RC} \text{ Hz}$$

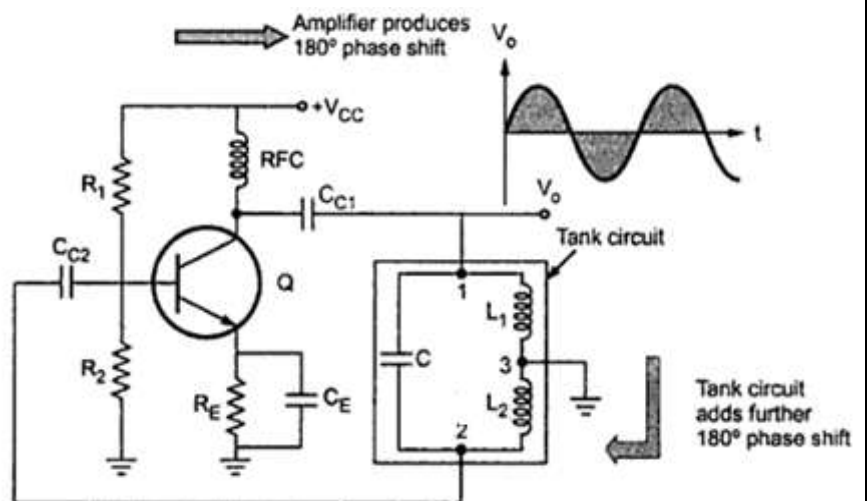
LC Tuned Oscillators

For generating frequencies beyond f_{unity} (1MHz), op-amp amplifiers can not be used. A properly biased BJT or FET amplifier along with a suitable feedback network is used to produce oscillations up to 500MHz.

Q14. Explain the operation of (a) Hartley Oscillator (b) Colpits Oscillator using BJT:

(a) Hartley Oscillator (BJT version)- A LC oscillator which uses two inductors and one capacitor in its feedback circuit along with an amplifier constitutes a Hartley Oscillator.

The amplifier uses an active device as a transistor in CE configuration. The circuit uses voltage divider bias provided by resistors R_1 and R_2 . A Radio frequency choke (RFC) is used in the collector circuit which acts as a short circuit under dc and an open circuit at high frequencies, hence provides isolation between dc and ac operations. The CE amplifier provides a phase shift of 180° ,



the feedback circuit another 180° phase shift. The feedback circuit consists of two series connected inductors L_1 and L_2 connected across the capacitor C forms the resonant circuit and Q of this tank circuit is usually greater than 10.

The output voltage appears across L_1 and the feedback voltage appears across L_2 . The feedback fraction in this type of oscillator is given by: $\beta = \left(\frac{L_2}{L_1}\right)$ and the gain of the amplifier $A \geq \left(\frac{L_2}{L_1}\right)$

Expression for frequency:

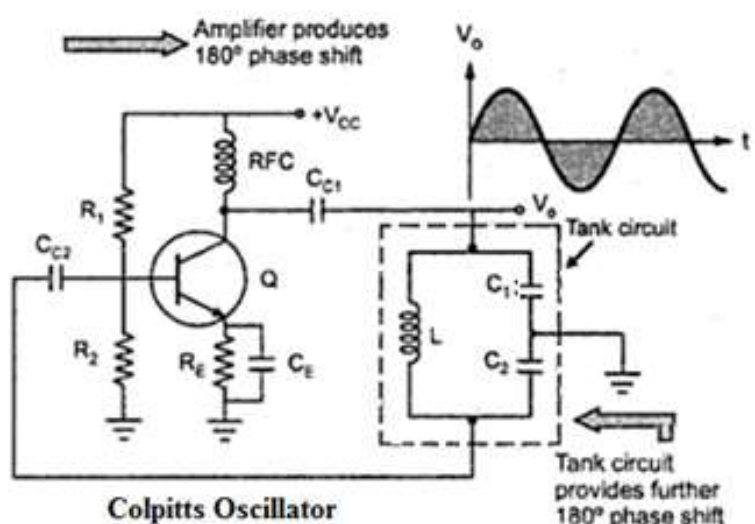
At the resonant frequency, the total reactance of the resonant circuit is zero and maximum voltage appears at the output and the resonant frequency is given by

$$f_r = \frac{1}{2\pi\sqrt{L_{eq}C}} \text{ Hz} \quad \text{where } L_{eq} = L_1 + L_2$$

(b) Colpitts Oscillator (BJT version)

A LC oscillator which uses two capacitors and one inductor in its feedback circuit constitutes a Colpitts Oscillator.

The amplifier uses an active device as a transistor in CE configuration. The circuit uses voltage divider bias provided by resistors R_1 and R_2 . A Radio frequency choke (RFC) is used in the collector circuit which acts as a short circuit under dc and an open circuit at high frequencies, hence provides isolation between dc and ac operations. The CE amplifier provides a phase shift of 180° , the feedback circuit gives another 180° phase shift. The feedback circuit consists of two series connected Capacitors C_1 and C_2 connected across the Inductor L forms the



resonant circuit and Q of this tank circuit is usually greater than 10.

The output voltage appears across C_1 and the feedback voltage appears across C_2 . The feedback fraction in this type of oscillator is given by: $\beta = \left(\frac{C_1}{C_2}\right)$ and the gain of the amplifier $A \geq \left(\frac{C_2}{C_1}\right)$

Expression for frequency:

At the resonant frequency, the total reactance of the resonant circuit is zero and maximum voltage appears at the output and the resonant frequency is given by

$$f_r = \frac{1}{2\pi\sqrt{LC_{eq}}} \text{ Hz} \quad \text{where } C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

Problem: For the transistor Colpitts oscillator circuit, calculate the oscillation frequency: if $L = 100 \text{ mH}$, $C_1 = 0.005 \text{ }\mu\text{F}$, $C_2 = 0.01 \text{ }\mu\text{F}$

$$\text{Solution: } C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{(0.005 \text{ }\mu\text{F})(0.01 \text{ }\mu\text{F})}{(0.005 + 0.01) \text{ }\mu\text{F}} = 3.3 \text{ nF}$$

$$f_o = \frac{1}{2\pi\sqrt{LC_{eq}}} = \frac{1}{2\pi\sqrt{100 \times 10^{-3} (3.3 \times 10^{-9})}} = 2.77 \text{ kHz}$$

Problem: Calculate the oscillator frequency for an FET Hartley oscillator $C = 250 \text{ pF}$, $L_1 = 1.5 \text{ mH}$, $L_2 = 1.5 \text{ mH}$.

$$\text{Solution: } L_{eq} = L_1 + L_2 = [1.5 + 1.5] \text{ mH} = 3 \text{ mH}$$

$$f_o = \frac{1}{2\pi\sqrt{L_{eq} C}} = \frac{1}{2\pi\sqrt{3 \times 10^{-3} (250 \times 10^{-12})}} = 184 \text{ kHz}$$

Problem: In a transistorized Hartley oscillator, the two inductances are 2 mH and $20 \text{ }\mu\text{H}$ while the frequency is to be changed from 950 kHz to 2050 kHz . Calculate the range over which the capacitor is to be varied.

$$\text{Solution: } L_{eq} = L_1 + L_2 = 2 \text{ mH} + 20 \text{ }\mu\text{H} = 2.02 \text{ mH}$$

$$f_{\max} = 2050 \text{ kHz} = \frac{1}{2\pi\sqrt{L_{eq} C_{\min}}} = \frac{1}{2\pi\sqrt{2.02 \times 10^{-3} C_{\min}}} \rightarrow C_{\min} = 2.98 \text{ pF}$$

$$f_{\min} = 950 \text{ kHz} = \frac{1}{2\pi\sqrt{L_{eq} C_{\max}}} = \frac{1}{2\pi\sqrt{2.02 \times 10^{-3} C_{\max}}} \rightarrow C_{\max} = 13.89 \text{ pF}$$

Hence C must be varied between 2.98 pF to 13.89 pF to get the required frequency variation.

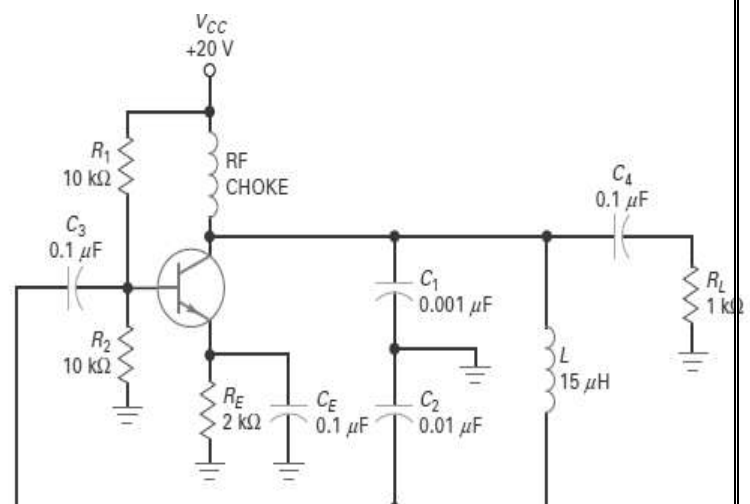
Problem- What is the frequency of oscillation in the circuit shown below? What is the feedback fraction?

How much voltage gain does the circuit need to start oscillating?

$$\text{Solution: } C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{(0.001 \text{ }\mu\text{F})(0.01 \text{ }\mu\text{F})}{(0.001 + 0.01) \text{ }\mu\text{F}} = 909 \text{ pF}$$

$$r = \frac{1}{2\pi\sqrt{LC_{eq}}} = \frac{1}{2\pi\sqrt{15 \times 10^{-6} (909 \times 10^{-12})}}$$

$$f_r = 1.36 \text{ MHz}$$



$$\beta = \left(\frac{C_1}{C_2}\right) = \frac{0,001}{0.01} = 0.1 \quad \text{and}$$

$$\text{gain of the amplifier } A_{V(min)} = \left(\frac{C_2}{C_1}\right) = \frac{0,01}{0.001} = 10$$

CRYSTAL OSCILLATOR

Q15. Write an explanatory note on Crystal Oscillator

A crystal oscillator is basically a tuned-circuit oscillator using a piezoelectric crystal as a resonant tank circuit. The crystal (usually quartz) has a greater frequency stability and are used in communication transmitters and receivers.

Characteristics of a Quartz Crystal

Piezo Electric Effect: Crystals like Quartz exhibits electro-mechanical resonance characteristics that when mechanical stress is applied across one set of its faces, a difference of potential develops across the opposite faces. This property of a crystal is called the piezoelectric effect. Similarly, a voltage applied across one set of faces of the crystal causes mechanical distortion in the crystal shape.

When alternating voltage is applied to a crystal, mechanical vibrations are set up—these vibrations having a natural resonant frequency dependent on the crystal. The crystal is represented by an equivalent electrical resonant circuit as shown in figure. The inductor L and capacitor C_s represent electrical equivalents of crystal mass and compliance, respectively, whereas resistance R is an electrical equivalent of the crystal structure's internal friction. The shunt capacitance C_m represents the capacitance due to mechanical mounting of the crystal. The equivalent Crystal's Q (quality factor) is very high—typically greater than 20,000..

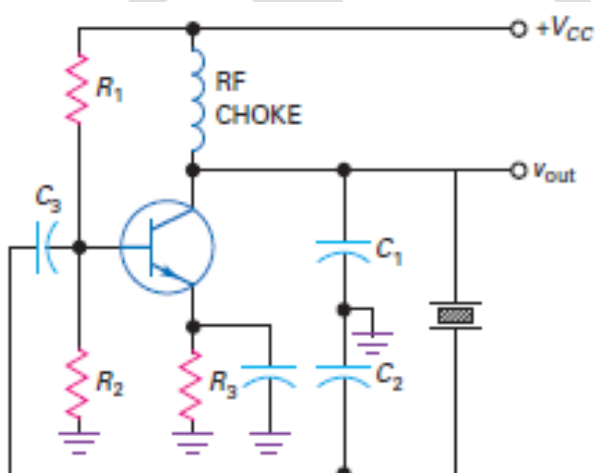
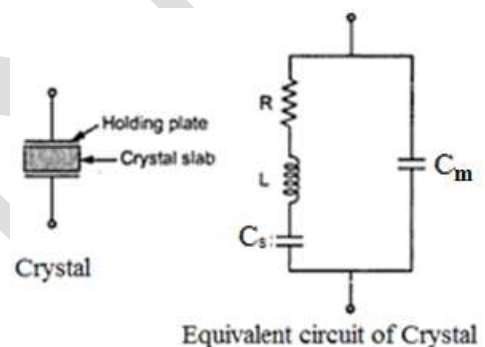


Figure shows a Crystal Oscillator circuit. The transistor operates in CE mode with voltage divider bias with proper operating point and the tank circuit is formed by series capacitors C_1 and C_2 in parallel with the crystal.

The series resonant frequency is given as

$$f_s = \frac{1}{2\pi\sqrt{L C_s}} \text{ Hz}$$

The parallel resonant frequency is given by

$$f_p = \frac{1}{2\pi\sqrt{L C_{eq}}} \text{ Hz} \quad \text{where } C_{eq} = \frac{C_p C_s}{C_p + C_s}$$

Applications of Crystal Oscillators:

The most important feature of crystal oscillator is its stability. Temperature compensated crystal oscillators (TXCO) are used in many digital IC's. Voltage controlled crystal oscillators (VCXO) are used in

PLL circuits where in the frequency is controlled by applied external voltage. Oven controlled crystal oscillators (most precise and highly stable) are used in wireless systems and base stations.

Problem : A crystal has a mounting capacitance of 10 pF. The inductance equivalent of mass is 1 mH, the frictional losses are 1 kΩ and the compliance is 1 pF. Find the series and parallel resonant frequency.

Solution: The series resonant frequency is given by

$$f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1 \times 10^{-3} (1 \times 10^{-12})}} = 5.033 \text{ MHz}$$

$$\text{The equivalent capacitance } C_{eq} = \frac{C_m C_s}{C_m + C_s} = \frac{10 (1)}{10 + 1} = 9.09 \text{ pF}$$

The parallel resonant frequency is given by

$$f_p = \frac{1}{2\pi\sqrt{LC_{eq}}} = \frac{1}{2\pi\sqrt{1 \times 10^{-3} (9.09 \times 10^{-12})}} = 5.275 \text{ MHz}$$

Problem: A 2 MHz quartz crystal is specified to have $L = 0.52\text{H}$, $C_s = 0.012\text{pF}$, $C_m = 4 \text{ pF}$ and $R = 120\Omega$.

Find f_o , f_s and f_p

Solution – $f_o = 2\text{MHz}$

$$f_s = \frac{1}{2\pi\sqrt{LC_s}} = \frac{1}{2\pi\sqrt{0.52 (0.012 \times 10^{-12})}} = 2.015 \text{ MHz}$$

$$\text{The equivalent capacitance } C_{eq} = \frac{C_p C_s}{C_p + C_s} = \frac{4 (0.012)}{4 + 0.012} = 0.01196 \text{ pF}$$

$$f_p = \frac{1}{2\pi\sqrt{LC_{eq}}} = \frac{1}{2\pi\sqrt{0.52 (0.01196 \times 10^{-12})}} = 2.018 \text{ MHz}$$

$$Q = \frac{\omega_0 L}{R} = \frac{2\pi f_o L}{R} = \frac{2\pi (2 \times 10^6) (0.52)}{120} = 54455$$

Comparison between LC oscillator and Crystal Oscillator

	LC oscillators	Crystal oscillator
1	Separate L and C components are necessary in the tuned circuit	The single crystal serves the purpose of tuned circuit
2	The Q Value of the tank circuit is less compared to crystal	The Q value is very high compared to LC circuits
3	The frequency stability is less	Very high frequency stability
4	The bandwidth of operation is more	Narrow band width
5	Changes in temperature affects the frequency	Frequency is independent of changes in temperature.
6	Used for general applications like in signal generators, FM tuners etc	Used for specific and precise applications like watches, computers, counters etc

Review Questions

1. What is an Oscillator? Explain the principle of sinusoidal oscillator.
2. State the conditions for generating sustained Oscillations. (Barkhausen's criterion)

3. With neat circuit diagram, explain the working of (i) Wien bridge oscillator using op-amp(ii) RC phase shift oscillator using op-amp (iii) Colpitt's Oscillator using BJT (iv) Hartley Oscillator using BJT (v) Crystal Oscillator using BJT. Write the expression for frequency.
4. Compare LC and Crystal Oscillators.

555 Timer & its Applications

Introduction: The 555 timer is a highly stable monolithic device for generating accurate time delays or oscillation. Signetics Corporation first introduced this device as SE555 / NE 555 and it is available in two package styles, 8 pin circular style TO-9 Can or 8 pin mini DIP. Also NE 556 which contains two 555 timers is available in 14 pin DIP.

Features: (i) The timer can be used with supply voltage in the range of + 5V to +18V and can drive a load up to 200 mA and is compatible to both TTL and CMOS logic circuits. (ii) It has a high temperature stability (0°C to 70°C) and operates in two modes as Monostable multivibrator and Astable multivibrator.(iii) Has an adjustable duty cycle, timing is from micro seconds through hours.

Applications: The timer can provide time delays ranging few micro seconds to hours. The various applications of timer includes oscillator, pulse generator, ramp, square wave and triangular wave generators, traffic light controls, burglar alarm, voltage monitors, automatic parking system of vehicles

IC 555 Timer Pins description:

Q16. Explain the pin configuration of 555 timer IC

Pin 1: Ground : All voltages are measured with respect to this terminal.

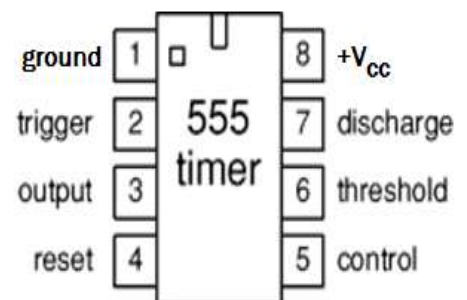
Pin 2: Trigger: This is the inverting terminal of the lower comparator, when the signal level at the trigger terminal goes below the reference level of the comparator ($\frac{V_{CC}}{3}$), the comparator output goes high and resets the flip-flop. Hence the output of the timer is low.

Pin 3: Output: The output terminal is connected to $\bar{Q}$ of the flip-flop through an output stage which has a power amplifier and an inverter. Hence when $\bar{Q} = 0$, the out will be high and $\bar{Q} = 1$, the output will be low.

Pin 4: Reset The reset input provides a mechanism to reset the flip-flop overriding the inputs. When not used, it is connected to V_{CC} to avoid false triggering.

Pin 5 : Control Voltage: Control voltage can be used to adjust the threshold voltage which is set internally to $(\frac{2}{3})V_{CC}$. By applying a voltage on this pin, the pulse width (duty cycle) of the output waveform can be varied (pulse width modulation).

Pin 6 : Threshold: This is the non-inv input terminal of the comparator 1, by applying signal more than the reference level of this comparator ($\frac{2}{3})V_{CC}$, the flip-flop can be set, hence the output of the timer goes high.

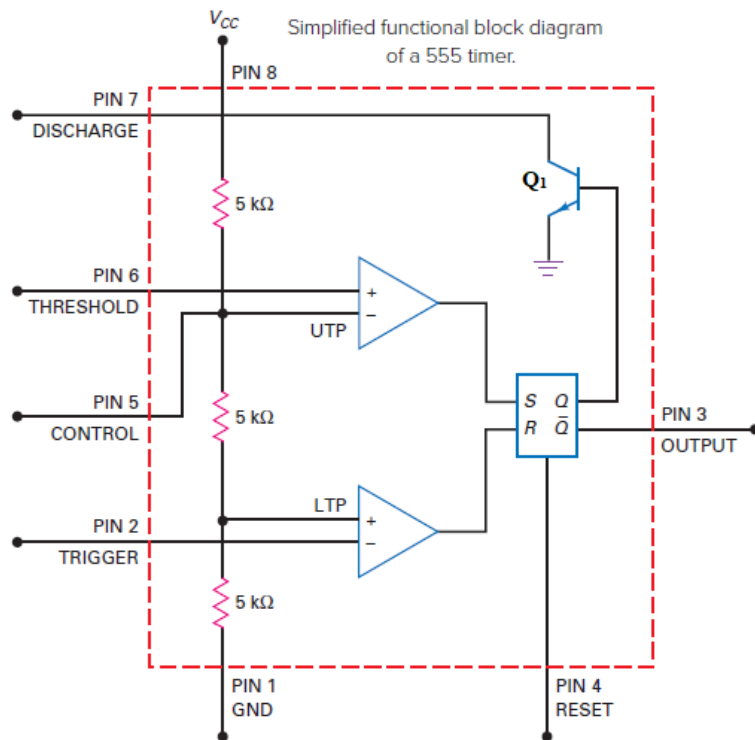


Pin 7 : Discharge This pin is connected to the collector terminal of the NPN transistor. This pin provides a path for charging and discharging for the external capacitor connected across it.

Pin 8 : Supply + V_{CC} : The 555 timer can work with any dc supply voltage in between + 4.5 V to + 15 V.

Functional Diagram and Description

Q16. With a neat functional diagram, explain the operation of 555 timer



Functional Diagram and Description

The 555 Timer consists of a potential divider network of three $5\text{ k}\Omega$ resistors, which provides a reference voltage levels of $\left(\frac{2}{3}\right)V_{CC}$ and $\left(\frac{1}{3}\right)V_{CC}$ to upper and lower comparators respectively and the output of these comparators will set or reset the SR flip-flop.

In the standby (stable) state, the output of the flip flop Q is high; hence the output at pin 3 is low. A negative going trigger pulse is applied to pin 2 and should have its dc level greater than $\left(\frac{1}{3}\right)V_{CC}$ (the threshold level of lower comparator). At the negative edge of the trigger

pulse, as the trigger passes through $\left(\frac{1}{3}\right)V_{CC}$, the output of the lower comparator goes high and will set the flip flop, hence $\bar{Q} = 1$ and $Q = 0$. Hence output is high

Similarly during positive excursion, when the threshold voltage at pin 6 exceeds $\left(\frac{2}{3}\right)V_{CC}$, the output of the upper comparator becomes high and it resets the flip-flop. ($Q = 1$ & $\bar{Q} = 0$). Output goes to low.

The reset input (pin4) provides a mechanism to reset the flip-flop over-riding the inputs. When not used, it is connected to V_{CC} . The transistor acts as a discharge transistor. When the output is high, Transistor is OFF making the discharge terminal (pin7) open. When the output is low, Q_1 is forward biased to ON condition. Then the Discharge terminal appears as a short circuit to ground.

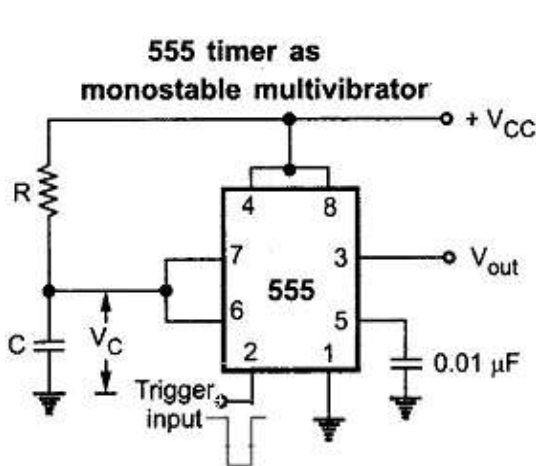
555 Timer as Monostable Multivibrator

With a neat functional diagram, circuit diagram and waveforms, explain the operation of Monostable multivibrator using 555 timer. Also derive the expression for Pulse width

A **Multivibrator** is a two-state circuit that has zero, one, or two stable output states.

A Monostable multivibrator has **only one stable state**. When trigger is applied, it produces a pulse at the output and returns back to its stable state.

The duration of the pulse depends on the values of external R and C .



Operation

Initially the output is low, hence the flip-flop output Q is high; this drives the transistor into saturation. The capacitor discharges completely and voltage across it is nearly zero. When a negative going trigger pulse is applied as input at pin 2 and when the trigger input falls to slightly less than $(1/3) V_{CC}$, the lower comparator resets the flip-flop ($Q = 0$ & $\bar{Q} = 1$). Since Q has changed to low, the transistor goes into cutoff, allowing the capacitor to charge. Hence capacitor starts charging through resistance R. The voltage across capacitor increases exponentially. The charging time depends upon the externally connected Resistor R and Capacitor C. When the capacitor voltage $V_C \geq \left(\frac{2}{3}\right) V_{CC}$; then comparator 1 output goes high. This makes the flip-flop output ($Q = 1$ & $\bar{Q} = 0$) and hence the out changes to low. The transistor is driven into saturation and hence capacitor voltage instantaneously discharges to zero (discharges through transistor).

Expression for Pulse Width:

The voltage across the capacitor is given by

$$V_C = V_{CC}(1 - e^{-t/RC})$$

$$\text{At } t = T_{ON}, V_C = \left(\frac{2}{3}\right) V_{CC}$$

$$\left(\frac{2}{3}\right) V_{CC} = V_{CC}(1 - e^{-T/RC}) \quad \text{or}$$

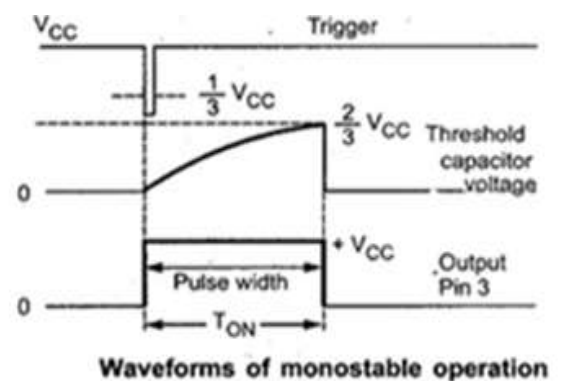
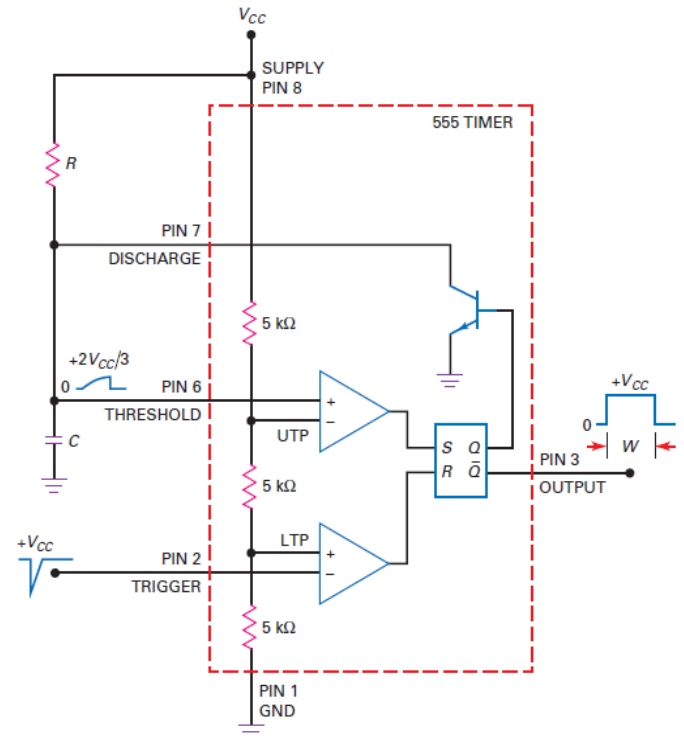
$$T_{ON} = -RC \ln(1/3) = 1.1 RC \text{ sec}$$

Hence the timing interval is independent of the supply voltage.

Problem: Design a Monostable multivibrator using 555 timer to obtain a pulse width of 10 msec.

Solution: $T = T_{ON} = 1.1 RC \rightarrow RC = \frac{T}{1.1} = \frac{10 \times 10^{-3}}{1.1} = 9.09 \text{ msec}$

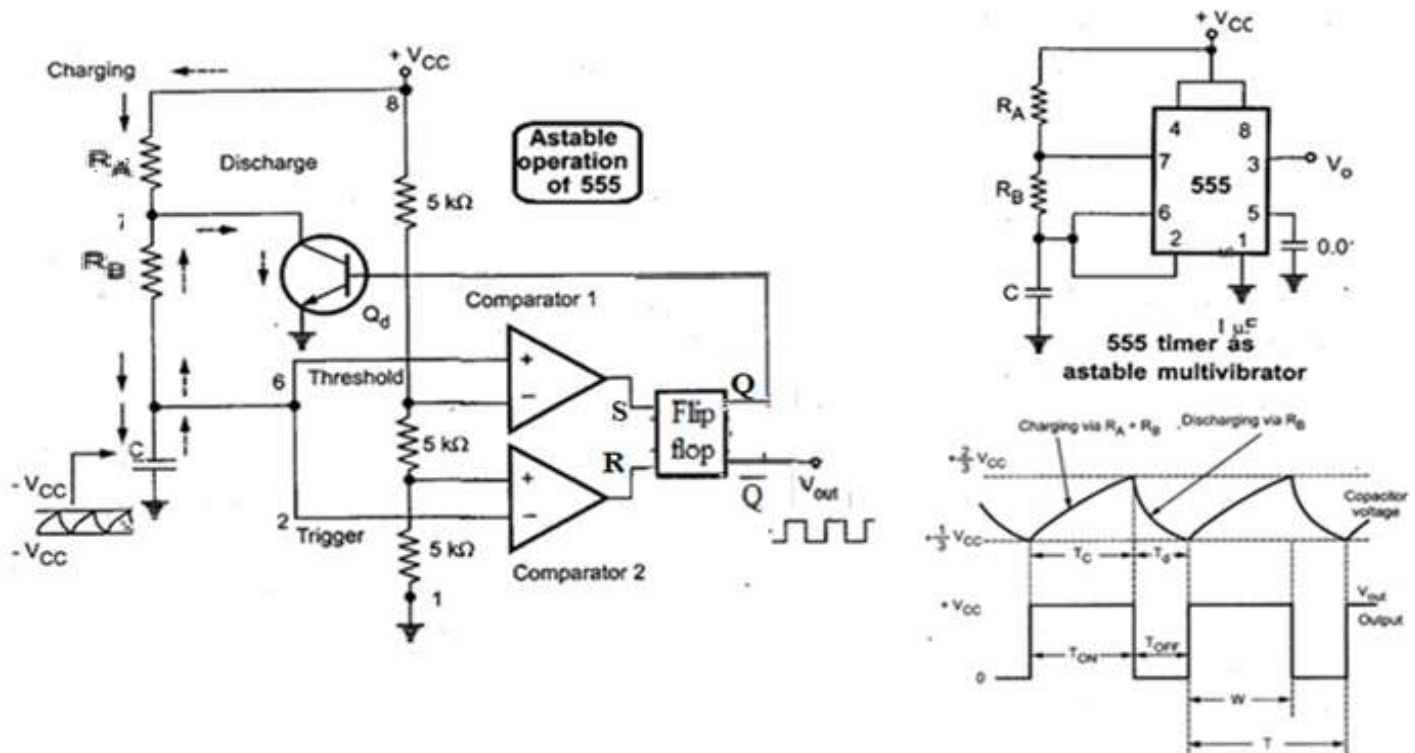
Let $C = 1 \mu F$, then $R = \frac{9.09 \times 10^{-3}}{1 \times 10^{-6}} = 9.09 \text{ k}\Omega$



555 Timer as Astable Multivibrator

With a neat functional diagram, circuit diagram and waveforms, explain the operation of Astable multivibrator using 555 timer. Also derive the expression for time period and duty cycle

An Astable multivibrator is also called as free running multivibrator, which does not have any stable states, is a rectangular wave generating circuit. This circuit does not require any external trigger to change the state of the output. Hence it is called as free running multivibrator.



Working: When the power supply is V_{CC} is connected to the circuit, the external timing capacitor C charges towards V_{CC} with a time constant $(R_1 + R_2)C$. During this time, output (pin 3) is held high (equal to V_{CC}) as the flip-flop is reset due to which $Q = 0$.

When the capacitor voltage is greater than or equal to $(2/3) V_{CC}$, the upper comparator output becomes high, which makes the flip-flop output $Q=1$, hence $\bar{Q} = 0$. The timer output at pin 3 goes to low and the transistor Q turns on. Now the capacitor C starts discharging towards ground potential through the resistor R_2 and transistor with a time constant R_2C . Once the capacitor voltage reaches $(1/3) V_{CC}$, the lower comparator triggers making the flip-flop output $Q = 0$ and $\bar{Q} = 1$, the transistor turns-off, making the output at pin 3 high. Now the capacitor starts charging towards V_{CC} and the process repeats.

Calculations:

The voltage across the capacitor subjected to step input V_{CC} is given by $V_c = V_{CC}(1 - e^{-t/RC})$

The time T_1 taken by the capacitor to charge from 0 to $(2/3) V_{CC}$ is given by $(\frac{2}{3}) V_{CC} = V_{CC} (1 - e^{-T_1/RC})$

$$T_1 = -RC \ln (1/3) = 1.1 RC \text{ sec}$$

The time T_2 taken by the capacitor to charge from 0 to $(1/3)V_{CC}$ is given by $(\frac{1}{3}) V_{CC} = V_{CC} (1 - e^{-T_2/RC})$

$$T_2 = -RC \ln (2/3) = 0.405RC \text{ sec}$$

Hence the time taken by the capacitor to charge from $(1/3) V_{CC}$ to $(2/3) V_{CC}$ is given by

$$T_C = T_{ON} = T_1 - T_2 = (1.09 - 0.405)RC = 0.69 RC$$

During charging both R_A and R_B are used, hence $T_{ON} = 0.693 (R_A + R_B) C$

The output is low while the capacitor discharges, the time taken by the capacitor to discharge from $(2/3) V_{CC}$ to $(1/3) V_{CC}$ is given by

$$\left(\frac{1}{3}\right) V_{CC} = \left(\frac{2}{3}\right) V_{CC} (1 - e^{-T_{OFF}/RC}) \rightarrow T_{OFF} = -RC \ln(1/2) = 0.693RC$$

Only R_B is used while discharging, hence $T_{OFF} = 0.693R_B C$

The total time $T = T_{ON} + T_{OFF} = 0.693 (R_A + 2R_B) C$ seconds

$$\text{The frequency } f = \frac{1}{T} = \frac{1}{0.693(R_A + 2R_B)C} = \frac{1.45}{(R_A + 2R_B)C} \text{ Hz}$$

Duty cycle is defined as the ratio of On time (T_C) to the total time T

$$\%D = \frac{T_{on}}{T} = \frac{T_{on}}{T_{on} + T_{loff}} = \frac{R_A + R_B}{R_A + 2R_B}$$

Note - A symmetrical Astable multivibrator for generating square wave can be constructed by connecting low resistance device like diode in the (forward bias path) charging path and keeping the resistances R_A and R_B values same, which ensures both charging and discharging times equal and gives 50% duty cycle.

Problem - A 555 timer Astable multivibrator has $R_1 = 2.2K\Omega$, $R_2 = 6.8 K\Omega$ and $C=0.01\mu F$. Calculate T_{high} , T_{low} , free running frequency and Duty cycle. Draw the circuit.

Solution: $T_{high} = 0.69(R_A + R_B) C = 0.69(2200+6800) \times 0.01 \times 10^{-6} = 0.062 \text{ m sec}$

$$T_{low} = 0.69 R_B C = 0.69(6800) \times 0.01 \times 10^{-6} = 0.047 \text{ m sec}$$

$$T = T_{high} + T_{low} = 0.109 \text{ m sec}$$

$$\text{Free running frequency } f = 1/T = 9.14 \text{ kHz} \quad \text{and} \quad \%D = \frac{T_{high}}{T} = \frac{T_{high}}{T_{high} + T_{low}} = \frac{0.062}{0.109} = 56\%$$

Design a square wave form generator of frequency 100 Hz and duty cycle 75%.

Solution: $T = \frac{1}{f} = \frac{1}{100} = 10 \text{ m sec}$

$$D = \frac{T_{high}}{T} \rightarrow T_{high} = D T = \frac{D}{f} = \frac{0.75}{100} = 7.5 \text{ msec}$$

$$T_{low} = T - T_{high} = 10 - 7.5 = 2.5 \text{ msec}$$

$$\text{Also } T_{low} = 0.69 R_2 C \quad \text{let } C = 1 \mu F, \text{ then } R_B = \frac{T_{low}}{0.69 C} = \frac{2.5 \times 10^{-3}}{0.69 \times 1 \times 10^{-6}} = 3.62 \text{ k}\Omega$$

7. Design a square wave generator of frequency 1 kHz with $D = 0.25$

Solution: $T = \frac{1}{f} = \frac{1}{1000} = 1 \text{ m sec} = 0.69(R_A + 2 R_B) C$

$$D = 0.25 = \frac{R_A + R_B}{R_A + 2R_B} \quad \text{or } (R_A + 2R_B) = 4 (R_A + R_B) \quad \text{or } R_B = 3 R_A$$

$$\text{Let } C = 0.1\mu F, \text{ then } T = 0.001 = 0.69(R_A + 3 R_A) \times 0.1 \times 10^{-6} \rightarrow R_A = 3.623 \text{ k}\Omega \text{ and } R_B \cong 12 \text{ k}\Omega$$